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Supplementary Materials for

Human brain glycoform coregulation network and glycan modification alterations in Alzheimer's disease

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The PDF file includes:

Figs. S1 to S11
Tables S1
Legends for tables S2 to S10

Other Supplementary Material for this manuscript includes the following:

Tables S2 to S10

SUPPLEMENTARY FIGURES

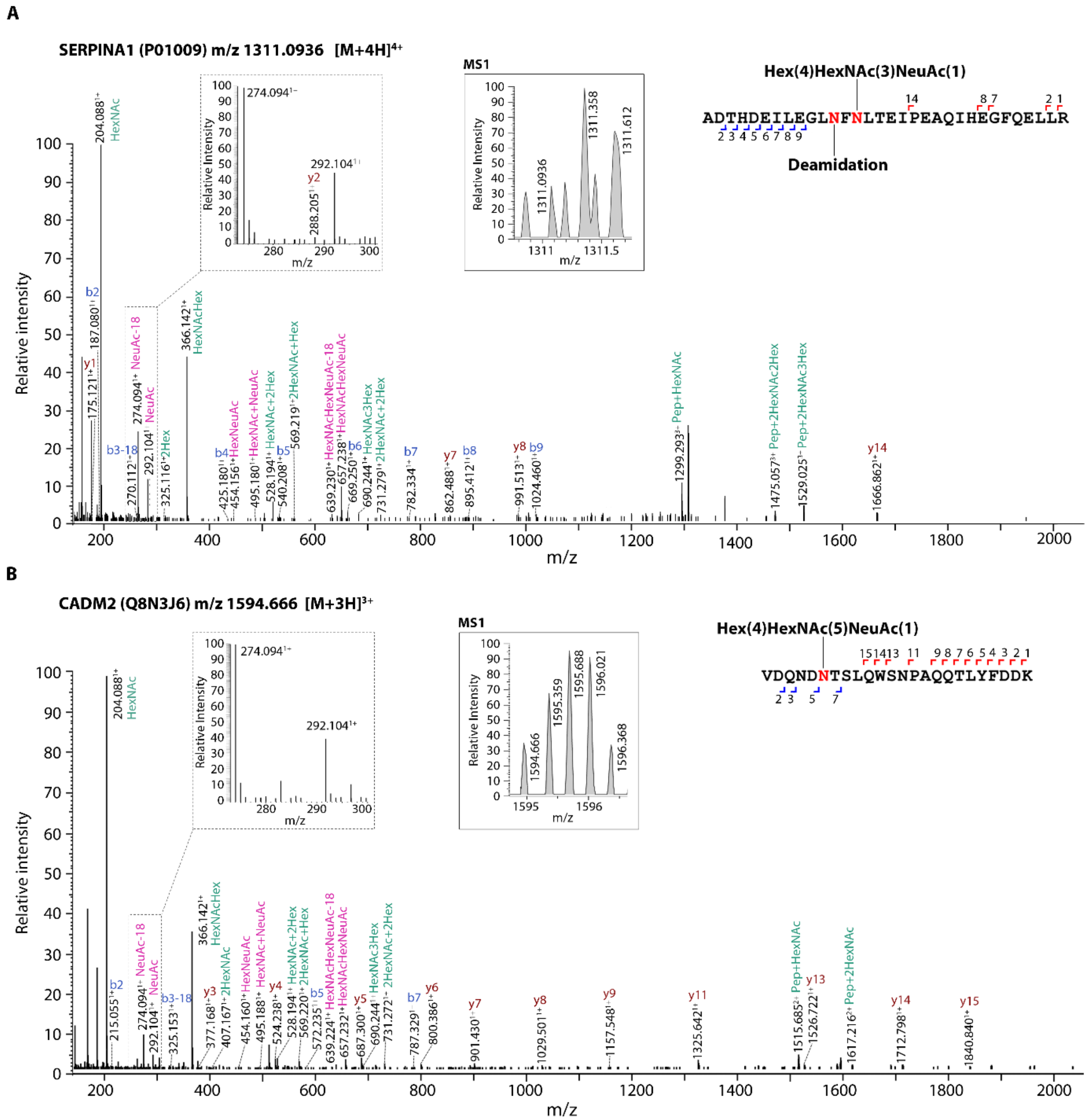

Fig. S1. Manual validation confirms Byonic annotation of NeuAc-containing glycopeptides. Representative SCE-HCD-MS/MS spectra of (**A**) a Byonic-identified sialylated glycopeptide containing both a deamidation and a NeuAc-containing glycan composition and (**B**) a Byonic-identified sialylated glycopeptide containing a NeuAc-containing glycan composition. Manual validation confirms that these spectra indeed correspond to the indicated NeuAc-containing *N*-glycopeptides from human SERPINA1 (A) and CADM2 (B) as shown by the presence of well-established diagnostic oxonium ions at m/z 292.10 for NeuAc and 274.09 for NeuAc with water loss (NeuAc-18) (see insert, dotted lines), NeuAc-containing glycan fragment ions (labeled in purple color), Y ions of the intact peptide with a fragmented glycan, and b- and y-ion peptide backbone fragmentation. The correct annotation of each spectrum is further supported by the monoisotopic precursor ion profile (see MS1 insert, solid lines).

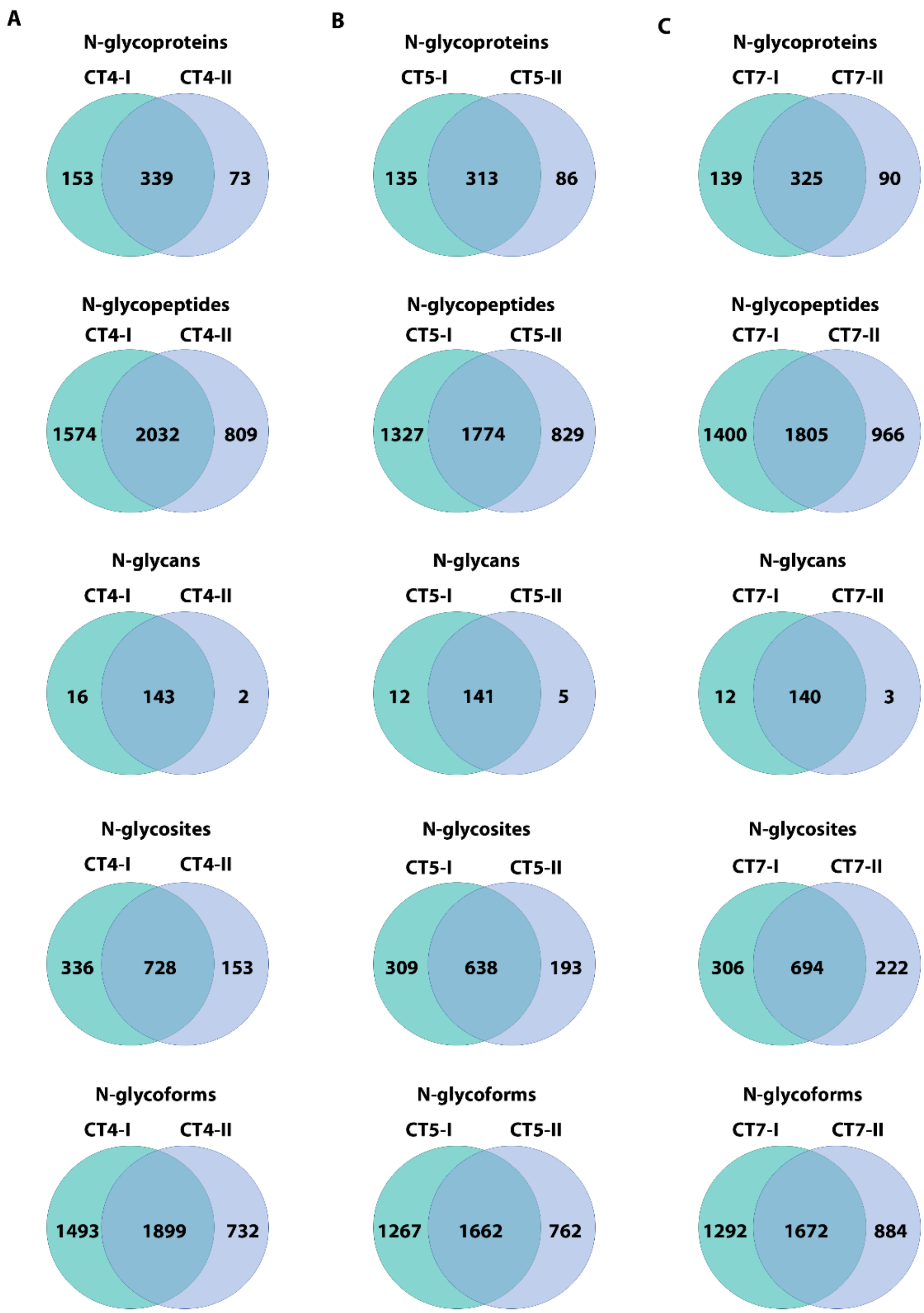

Fig. S2. Glycoproteomics analysis of technical replicates from control brain samples. Venn diagrams showing the overlap of N-glycoproteins, N-glycopeptides, N-glycans, N-glycosites, and N-glycoforms identified in technical replicates from three control brain samples, CT4, CT5, and CT7. Technical replicates are indicated as sample numbers I and II.

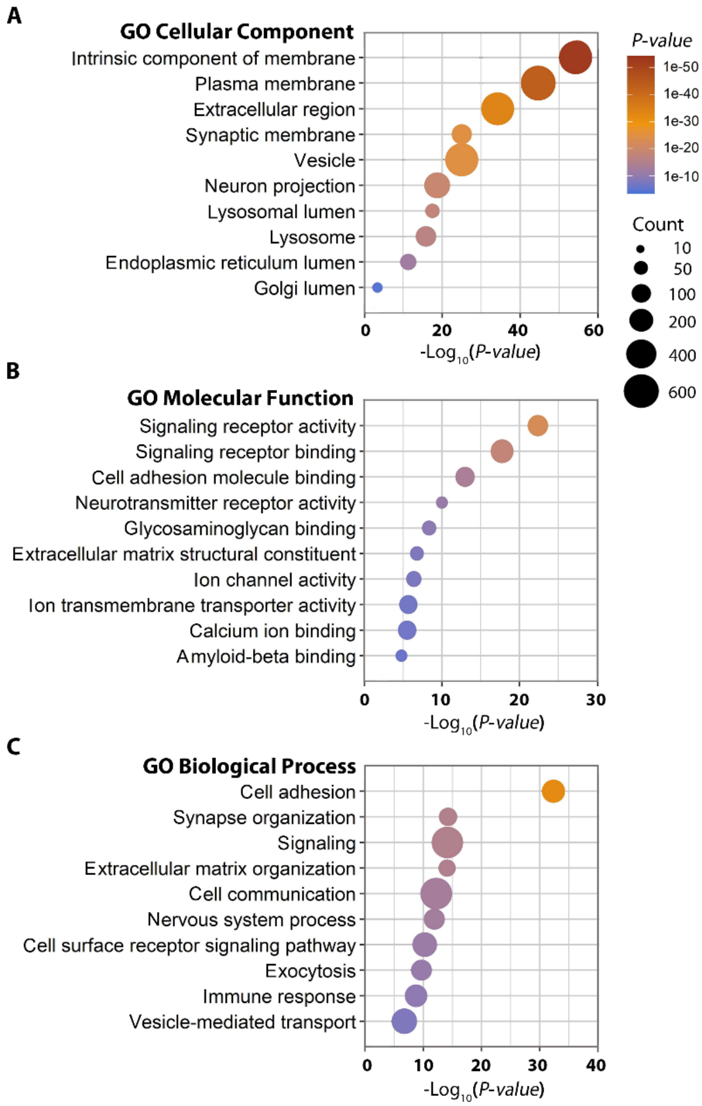

Fig. S3. Gene ontology enrichment analysis of human brain N-glycoproteins identified by intact glycoproteomics. GO cellular component (**A**), molecular function (**B**), and biological process (**C**) enriched in the intact glycoproteome dataset are shown with Benjamini-Hochberg FDR-corrected *P* values. Count indicates the number of glycoproteins per GO term.

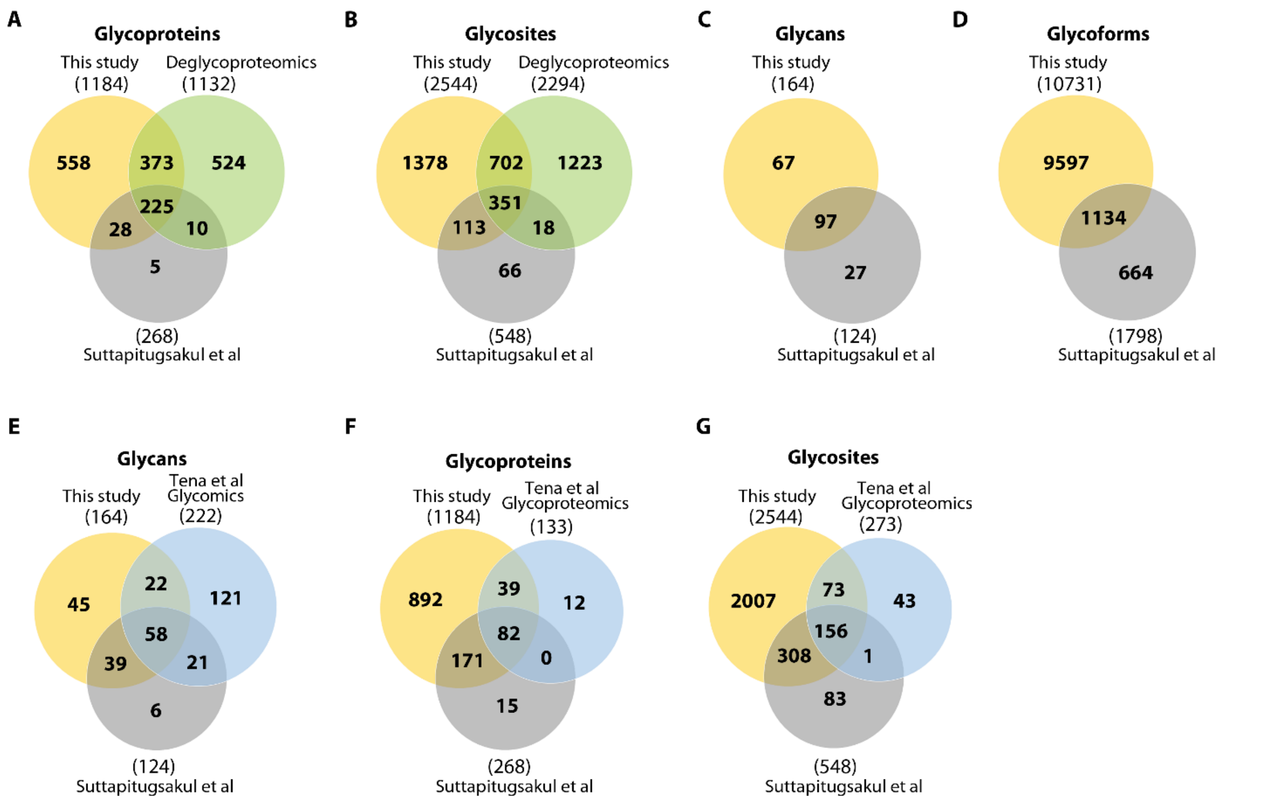

Fig. S4. Comparison of human brain glycoproteomics dataset obtained in this study to Suttapitugsakul et al and Tena et al datasets and our deglycoproteomics dataset. (A to **D)** Venn diagrams showing the overlap of N-glycoproteins, N-glycosites, N-glycans, and N-glycoforms identified in this study with those from Suttapitugsakul et al (*Ref. 24*) and our deglycoproteomics study (*Ref. 23*). (**E** to **G**) Venn diagrams showing the overlap of N-glycans, N-glycoproteins, and N-glycosites identified in dorsolateral prefrontal cortex by this study with those identified in membrane fractions of human frontal cortex samples by glycomics (E) and glycoproteomics (F and G) analyses from Tena et al (*Ref. 32*) and Suttapitugsakul et al (*Ref. 24*).

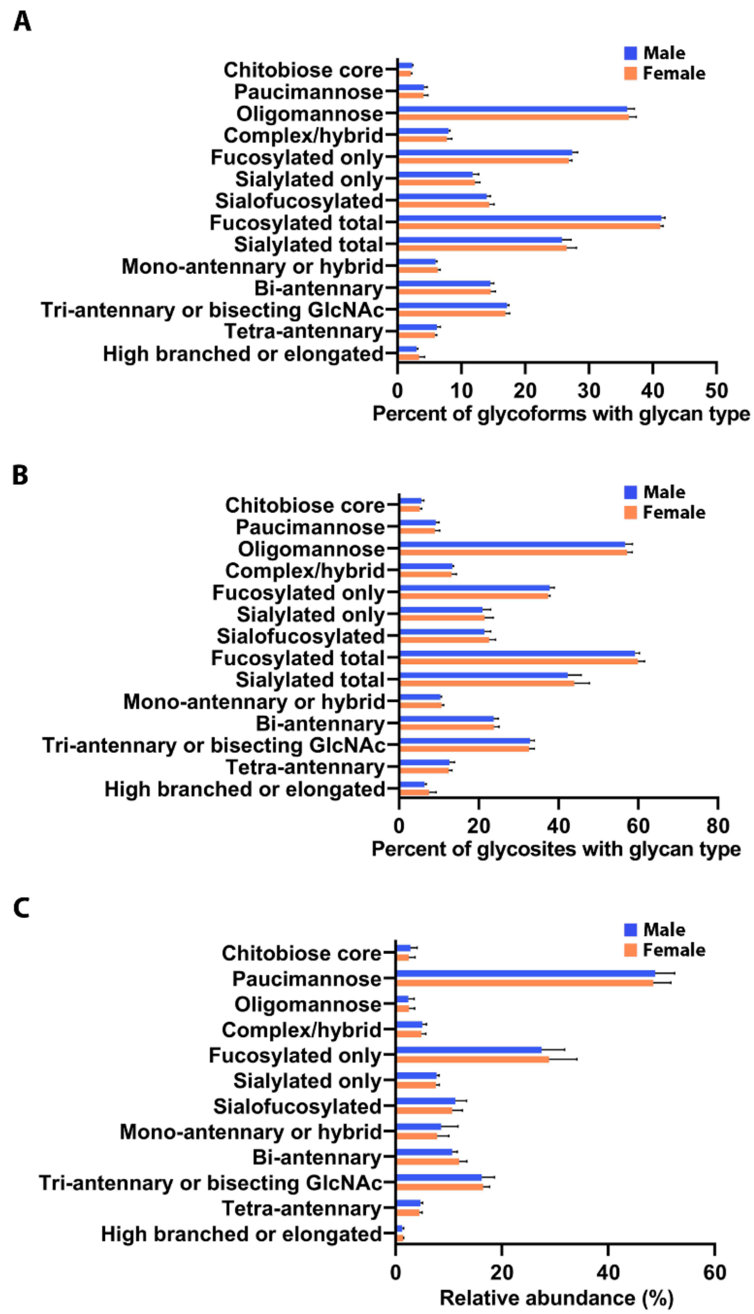

Fig. S5. Comparison of human brain N-glycosylation profiles shows no significant difference between sexes. (**A** and **B**) Comparison of the distribution of N-glycoforms (A) and N-glycosites (B) containing the indicated glycan type in male versus female control brains. (**C**) Comparison of relative abundances of glycan modifications by the indicated glycan types in male versus female control brains. Data represent means $\pm$ SD (n = 4 subjects per sex group). Unpaired two-tailed Student's t test shows no significant difference between sexes in the distribution or abundances for each glycan type.

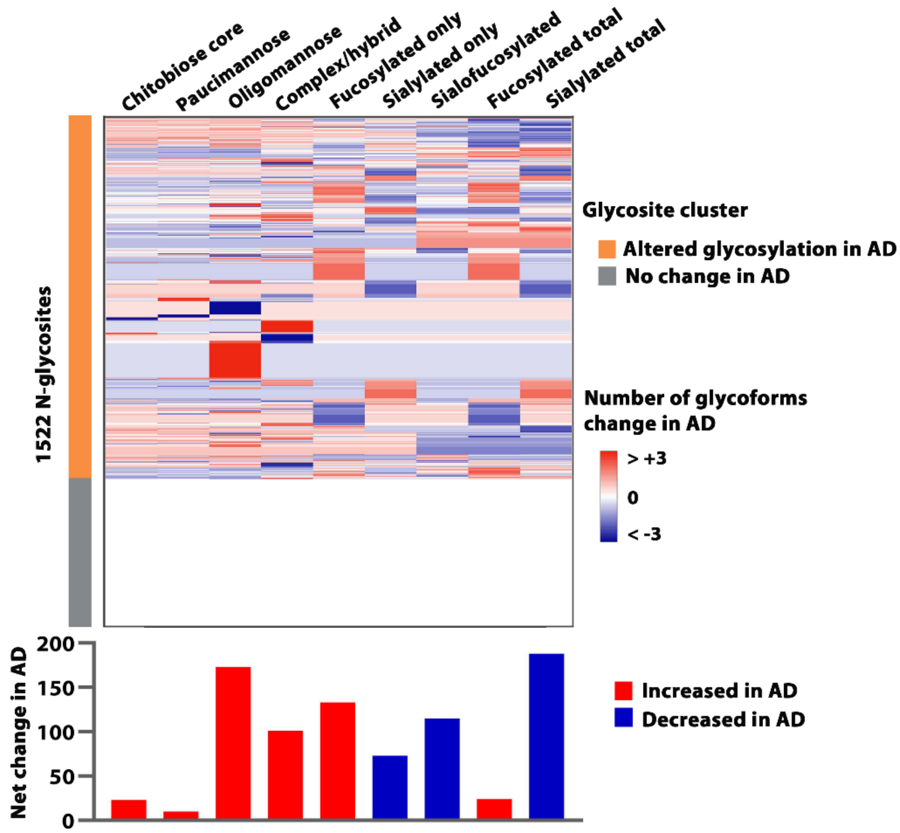

Fig. S6. Disease-associated changes in glycosylation patterns of 1522 N-glycosites identified in both control and AD brains. Heatmap (*top*) shows the unsupervised hierarchical clustering of 1522 N-glycosites based on site-specific changes in the number of glycoforms in AD versus the control for each of the indicated glycan types. The net change in the number of glycoforms for each glycan type across the 1522 glycosites in AD versus the control is shown at the *bottom*.

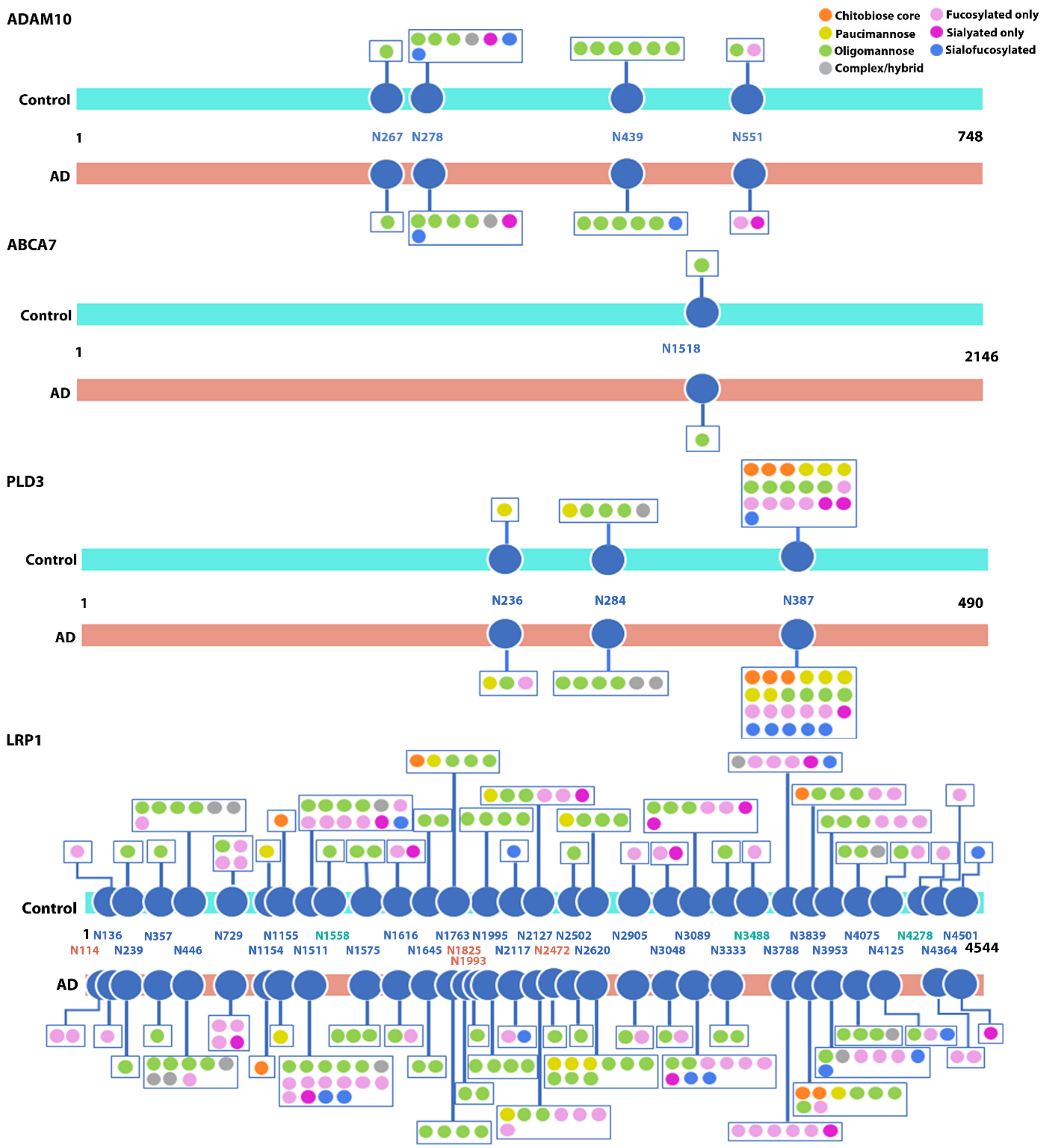

Fig. S7. Site-specific glycosylation patterns of ADAM10, ABCA7, PLD3, and LRP1 in AD and control brains. N-glycosites are indicated by the amino acid positions of glycosylated asparagine residues, with cyan- and pink-colored glycosites denoting the N-glycosites detected exclusively in control and AD brains, respectively.

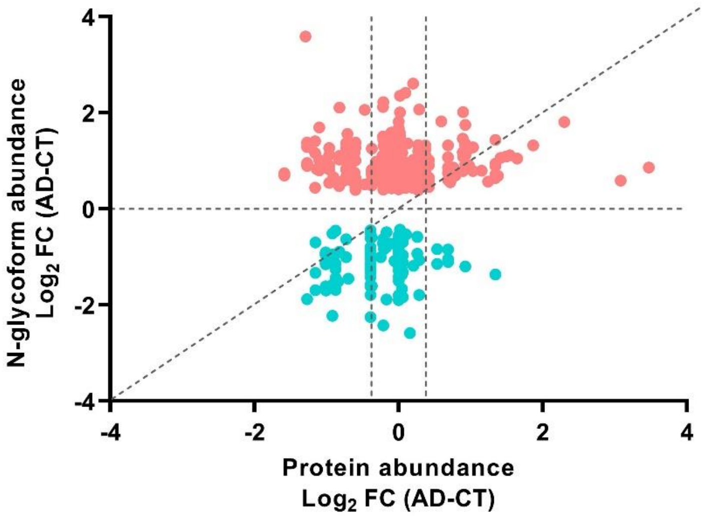

Fig. S8. Comparison of glycoform abundance changes in AD with corresponding protein abundance changes. Scatter plot showing that glycoform abundance changes of many AD-associated N-glycoforms are not due to altered protein abundance.

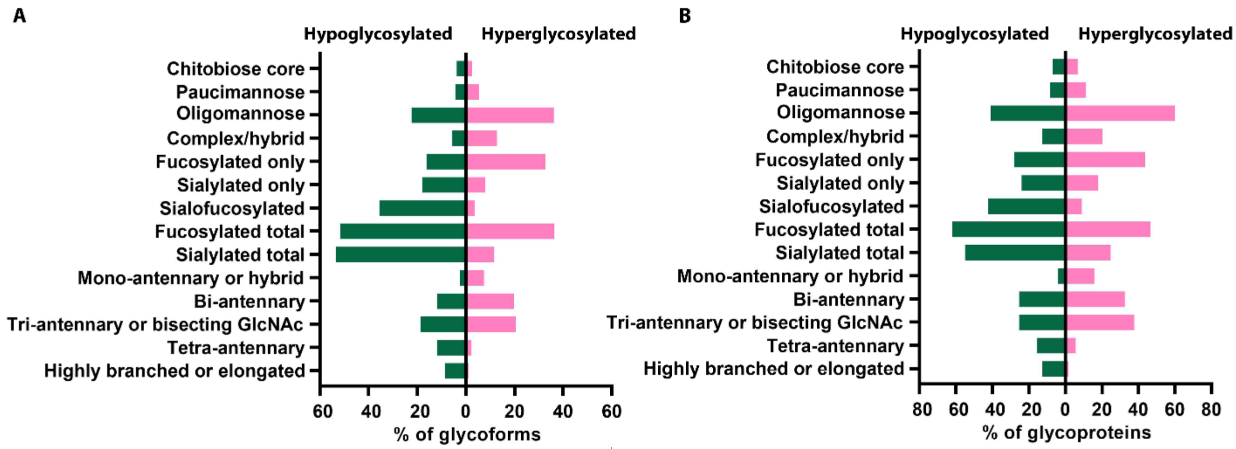

Fig. S9. Distribution of glycoforms and glycoproteins by glycan type in AD hyperglycosylation and hypoglycosylation datasets. Bar graphs showing the percentages of AD-associated hypoglycosylated or hyperglycosylated glycoforms (A) and glycoproteins (B) carrying the indicated glycan types.

Glycan module-trait relationships

	Age	Sex	AD status	CERAD	Braak	ApoE	PMI
GNM8	0.23 (0.4)	0.27 (0.3)	-0.22 (0.4)	-0.32 (0.2)	-0.15 (0.6)	-0.097 (0.7)	0.25 (0.4)
GNM4	-0.14 (0.6)	-0.14 (0.6)	-0.52 (0.04)	-0.57 (0.02)	-0.45 (0.08)	-0.25 (0.4)	0.0057 (1)
GNM3	0.041 (0.9)	0.16 (0.6)	-0.38 (0.1)	-0.4 (0.1)	-0.42 (0.1)	0.31 (0.2)	0.5 (0.056)
GNM1	-0.0014 (1)	0.23 (0.4)	-0.9 ($3x10^{-6}$)	-0.87 ($1x10^{-5}$)	-0.9 ($2x10^{-6}$)	-0.24 (0.4)	0.31 (0.2)
GNM7	0.073 (0.8)	-0.0032 (1)	0.13 (0.6)	0.15 (0.6)	0.042 (0.9)	-0.071 (0.8)	0.28 (0.3)
GNM5	-0.073 (0.8)	0.35 (0.2)	0.14 (0.6)	0.2 (0.5)	0.047 (0.9)	0.26 (0.3)	0.36 (0.2)
GNM2	0.042 (0.9)	-0.28 (0.3)	0.92 ($3x10^{-7}$)	0.91 ($1x10^{-6}$)	0.85 ($4x10^{-5}$)	0.29 (0.3)	-0.15 (0.6)
GNM6	-0.36 (0.2)	-0.16 (0.6)	0.24 (0.4)	0.25 (0.4)	0.23 (0.4)	-0.19 (0.5)	-0.021 (0.9)

Fig. S10. Module-trait relationships of glycan network modules to AD-related traits or sample characteristics. Biweight midcorrelations between module eigenglycans and the indicated trait are shown with the correlation coefficients (bicor *r*-values) on the top and *P* values in the bracket below. Significant positive correlations ($r > 0.50$, $P < 0.05$) are highlighted in *Red*, and significant negative correlations ($r < -0.50$, $P < 0.05$) in *Green*. PMI, postmortem interval.

Glycoform module-trait relationships

	Age	Sex	AD status	CERAD	Braak	ApoE	PMI
GFM3	0.071 (0.08)	0.2 (0.5)	-0.92 ($4x10^{-7}$)	-0.88 ($7x10^{-6}$)	-0.91 ($1x10^{-6}$)	-0.25 (0.4)	0.24 (0.4)
GFM10	0.48 (0.06)	-0.26 (0.3)	-0.37 (0.2)	-0.39 (0.1)	-0.47 (0.07)	0.28 (0.3)	-0.035 (0.9)
GFM13	0.47 (0.1)	0.11 (0.7)	-0.39 (0.1)	-0.34 (0.2)	-0.49 (0.06)	0.088 (0.7)	0.13 (0.6)
GFM17	0.43 (0.1)	0.15 (0.6)	-0.12 (0.7)	-0.13 (0.6)	-0.21 (0.4)	0.38 (0.1)	0.36 (0.2)
GFM14	0.4 (0.1)	0.46 (0.07)	-0.68 (0.004)	-0.66 (0.005)	-0.73 (0.001)	-0.19 (0.5)	0.45 (0.08)
GFM20	0.41 (0.1)	0.54 (0.03)	-0.028 (0.9)	-0.1 (0.7)	-0.073 (0.8)	0.0055 (1)	0.31 (0.2)
GFM9	0.14 (0.6)	0.38 (0.1)	0.24 (0.4)	0.17 (0.5)	0.29 (0.3)	0.07 (0.8)	0.15 (0.6)
GFM15	-0.11 (0.7)	-0.093 (0.7)	0.29 (0.3)	0.26 (0.3)	0.25 (0.4)	0.21 (0.4)	-0.19 (0.5)
GFM16	0.39 (0.1)	0.2 (0.5)	-0.18 (0.5)	-0.16 (0.6)	-0.2 (0.4)	0.25 (0.4)	0.29 (0.3)
GFM7	0.3 (0.3)	0.17 (0.5)	-0.18 (0.5)	-0.11 (0.7)	-0.27 (0.3)	-0.058 (0.8)	0.14 (0.6)
GFM5	0.058 (0.8)	0.058 (0.8)	0.17 (0.5)	0.18 (0.5)	0.1 (0.7)	-0.25 (0.3)	-0.13 (0.6)
GFM21	0.055 (0.8)	-0.13 (0.6)	0.16 (0.5)	0.23 (0.4)	0.029 (0.9)	0.0089 (1)	-0.38 (0.1)
GFM2	-0.091 (0.7)	-0.27 (0.3)	0.62 (0.01)	0.68 (0.003)	0.48 (0.06)	0.43 (0.1)	-0.14 (0.6)
GFM8	-0.027 (0.9)	0.051 (0.9)	0.42 (0.1)	0.41 (0.1)	0.3 (0.3)	0.22 (0.4)	-0.093 (0.7)
GFM19	-0.45 (0.08)	-0.17 (0.5)	0.72 (0.002)	0.72 (0.002)	0.63 (0.006)	-0.035 (0.9)	-0.31 (0.2)
GFM4	-0.34 (0.2)	-0.43 (0.09)	0.7 (0.002)	0.67 (0.004)	0.66 (0.006)	-0.1 (0.7)	-0.45 (0.08)
GFM12	0.14 (0.6)	-0.078 (0.8)	-0.23 (0.4)	-0.34 (0.2)	-0.16 (0.5)	$8x10^{-4}$ (1)	0.15 (0.6)
GFM18	0.4 (0.1)	-0.3 (0.3)	-0.12 (0.7)	-0.16 (0.6)	-0.19 (0.5)	0.41 (0.1)	0.24 (0.4)
GFM1	-0.12 (0.7)	-0.33 (0.2)	0.9 ($2x10^{-6}$)	0.86 ($2x10^{-5}$)	0.84 ($4x10^{-5}$)	0.53 (0.03)	-0.16 (0.6)
GFM11	0.016 (1)	-0.45 (0.08)	0.52 (0.04)	0.43 (0.1)	0.53 (0.04)	0.56 (0.02)	0.074 (0.8)
GFM6	0.34 (0.2)	-0.2 (0.5)	0.27 (0.3)	0.21 (0.4)	0.22 (0.4)	0.64 (0.008)	0.22 (0.4)

Fig. S11. Module-trait relationships of protein glycoform network modules to AD-related traits or sample characteristics. Biweight midcorrelations between module eigenglycoforms and the indicated trait are shown by bicor *r*-values on the top and *P* values in the bracket below. Significant positive ($r > 0.50$, $P < 0.05$) or negative ($r < -0.50$, $P < 0.05$) correlations are highlighted. PMI, postmortem interval.

SUPPLEMENTARY TABLES

Table S1. Case information of human AD and control brain samples

Case	Sex	Age at death (yr)	Age at onset (yr)	Disease duration (yr)	Braak stage	CERAD score	ApoE genotype	PMI (hr)
Control								
CT1	Male	65			0	Sparse	E3/3	6
CT2	Female	75			I	None	E3/3	6
CT3	Male	61			II	None	E3/4	<12
CT4	Female	74			II	None	E3/3	7
CT5	Male	59			I	None	E2/3	6
CT6	Female	78			II	None	E3/3	11.5
CT7	Male	94			II	None	E3/3	5.5
CT8	Female	61			II	None	n.d.	6
Mean ± SEM		70.88 ± 4.19						7.50 ± 0.94
Alzheimer's disease								
AD1	Male	79	73	6	V-VI	Frequent	E4/4	6
AD2	Female	72	59	13	VI	Frequent	E3/4	7
AD3	Male	67	56	11	VI	Frequent	E2/3	6.5
AD4	Male	77	70	7	VI	Frequent	E3/4	12
AD5	Male	74	60	14	VI	Frequent	E3/3	2.5
AD6	Male	68	60	8	VI	Frequent	E3/3	14.5
AD7	Female	61	51	10	VI	Frequent	E3/3	6
AD8	Male	69	59	10	VI	Frequent	E3/4	3.5
Mean ± SEM		70.88 ± 2.07						7.25 ± 1.44

Supplementary Tables S2 to S10 are provided in separate Microsoft Excel files.

Captions for Tables S2 to S10

Table S2 (Microsoft Excel format). Intact N-glycopeptides, N-glycoproteins, N-glycosites, N-glycans, and N-glycoforms identified in AD and control brains.

Table S3 (Microsoft Excel format). Gene ontology (GO) enrichment analysis of human brain glycoproteins identified by intact glycoproteomics.

Table S4 (Microsoft Excel format). Quantitative analysis of N-glycan modification levels by glycan type (A) or glycan composition (B) in AD and controls and identified glycan compositions with altered N-glycan modification levels in AD brain (C).

Table S5 (Microsoft Excel format). Differential glycoform abundance analysis (A) and identified glycoforms with altered site-specific N-glycan modification levels (B) or with a complete loss or gain of N-glycan modification (C) in AD.

Table S6 (Microsoft Excel format). Identified glycoforms and glycoproteins with hyperglycosylation (A), hypoglycosylation (B), hyperoligomannosylation (C), hypo-oligomannosylation (D), hyperpaucimannosylation (E), hypopaucimannosylation (F), hyperfucosylation (G), hypofucosylation (H), hypersialylation (I), or hyposialylation (J) in AD.

Table S7 (Microsoft Excel format). GO enrichment analysis of glycoproteins with hyperglycosylation (A), hypoglycosylation (B), hyperoligomannosylation (C), hypo-oligomannosylation (D), hyperpaucimannosylation (E), hypopaucimannosylation (F), hyperfucosylation (G), hypofucosylation (H), hypersialylation (I), or hyposialylation (J) in AD.

Table S8 (Microsoft Excel format). Glycan modification co-regulation network analysis.

Table S9 (Microsoft Excel format). Protein glycoform co-regulation network analysis.

Table S10 (Microsoft Excel format). GO enrichment analysis of glycoproteins in AD-associated glycoform modules.